

Module 10.2: Enterprise Resource Planning

Contents

ERP Market	1
ERP Marketplace.....	1
SAP AG.....	2
PeopleSoft	2
BAAN	2
JD Edwards.....	3
Oracle Corporation.....	3
ERP – Present and Future.....	3
Enterprise Application Integration (EAI)	4
ERP and E-Commerce	4
ERP and Internet	4
Future Directions in ERP	5

ERP Market

The ERP market consists of software vendors that develop and provide ERP solutions to organizations for managing business processes such as finance, production, sales, human resources, and supply chain management. ERP vendors offer integrated software packages that help businesses improve efficiency, productivity, and decision-making. **Features of the ERP Market**

1. Provides integrated business management solutions.
2. Supports organizations of different sizes and industries.
3. Offers both on-premise and cloud-based ERP systems.
4. Includes industry-specific ERP applications.
5. Continuously evolves with new technologies and business requirements.

ERP Marketplace

The ERP Marketplace refers to the global market where ERP vendors, consultants, implementation partners, and customers interact for the development, sale, implementation, and support of ERP systems.

Characteristics of ERP Marketplace

1. Presence of multiple ERP vendors.
2. Availability of customized ERP solutions.
3. Competition based on features, cost, and support.
4. Growing demand for cloud ERP systems.

5. Support services such as training and maintenance.

Benefits

1. Wide range of ERP options.
2. Better business process integration.
3. Access to advanced technologies.
4. Improved operational efficiency.
5. Enhanced business competitiveness.

SAP AG

SAP AG is one of the world's leading ERP software companies. SAP stands for **Systems, Applications, and Products in Data Processing**. Founded in Germany in 1972, SAP provides comprehensive ERP solutions for businesses of all sizes. **Features of SAP**

1. Integrated business management system.
 2. Real-time data processing.
 3. Strong financial management capabilities.
 4. Advanced reporting and analytics.
 5. Supply chain and production management support.
 6. Cloud and on-premise deployment options.
- Finance (FI)
 - Controlling (CO)
 - Materials Management (MM)
 - Sales and Distribution (SD)
 - Human Resources (HR)
 - Production Planning (PP)

Advantages

1. Highly scalable.
2. Supports global business operations.
3. Strong security and reliability.
4. Comprehensive business functionality.

Major SAP Modules

PeopleSoft

PeopleSoft is an ERP software suite originally developed by PeopleSoft Inc. and later acquired by Oracle Corporation. It is particularly known for its Human Resource Management capabilities.

Features of PeopleSoft

1. Human Resource Management.
 2. Financial Management.
 3. Supply Chain Management.
 4. Customer Relationship Management.
 5. Payroll and Workforce Management.
 6. Web-based user interface.
2. User-friendly interface.
 3. Flexible business processes.
 4. Good reporting capabilities.

Applications

- Employee management
- Payroll processing
- Recruitment
- Financial operations
- Workforce planning

Advantages

1. Strong HR functionality.

BAAN

BAAN is an ERP software package originally developed by Baan Company in the Netherlands. It became popular among manufacturing and industrial organizations due to its strong production management capabilities. **Features of BAAN**

1. Manufacturing management.
 2. Supply chain management.
 3. Financial management.
 4. Project management.
 5. Distribution management.
 6. Enterprise integration.
2. Flexible implementation.
 3. Efficient production planning.
 4. Better inventory control.

Applications

- Manufacturing industries
- Engineering companies
- Industrial production
- Supply chain operations

Advantages

1. Strong manufacturing support.

JD Edwards

JD Edwards is an ERP software solution that was acquired by Oracle Corporation. It is designed to support medium and large organizations across various industries. **Features of JD Edwards**

1. Financial management.
2. Human capital management.
3. Supply chain management.
4. Asset lifecycle management.
5. Project management.
6. Manufacturing management.
3. Strong reporting tools.
4. Scalable architecture.

Applications

- Manufacturing
- Construction
- Retail
- Distribution
- Service industries

Advantages

1. Industry-specific solutions.
2. Flexible deployment options.

Oracle Corporation

Oracle Corporation is a leading global software company that provides ERP, database, cloud computing, and enterprise software solutions. Oracle ERP systems are widely used by large organizations worldwide. **Features of Oracle ERP**

1. Integrated enterprise applications.
2. Cloud-based ERP solutions.
3. Financial and accounting management.
4. Human resource management.
5. Procurement and supply chain management.
6. Advanced analytics and reporting.
- Human Capital Management (HCM)
- Project Management
- Supply Chain Management
- Customer Experience (CX)

Advantages

1. High scalability.
2. Strong database integration.
3. Advanced security features.
4. Real-time business intelligence.
5. Global business support.

Major Oracle ERP Modules

- Financial Management
- Procurement

Comparison of Major ERP Vendors

Vendor	Strength
SAP AG	Comprehensive enterprise-wide integration
PeopleSoft	Human Resource and Workforce Management
BAAN	Manufacturing and Production Management
JD Edwards	Industry-specific ERP solutions
Oracle Corporation	Cloud ERP and Database Integration

These ERP vendors have played a significant role in the growth and development of modern Enterprise Resource Planning systems and continue to provide solutions for organizations across different industries.

ERP – Present and Future

ERP systems have evolved from simple business management software into comprehensive enterprise solutions that integrate business processes, support e-commerce, and utilize internet technologies.

Modern ERP systems focus on real-time information sharing, cloud computing, artificial intelligence, and global business integration.

Enterprise Application Integration (EAI)

Enterprise Application Integration (EAI) is the process of connecting different software applications within an organization so that they can share data and work together seamlessly.

EAI helps ERP systems communicate with other business applications such as CRM, SCM, HRM, and e-commerce platforms.

Objectives of EAI

1. Integrate different enterprise applications.
2. Improve data sharing across departments.
3. Eliminate duplicate data entry.
4. Increase operational efficiency.
5. Support real-time information exchange.
2. Real-time data exchange.
3. Business process integration.
4. Data synchronization.
5. Centralized information access.

Benefits of EAI

1. Improved communication between systems.
2. Faster business processes.
3. Reduced operational costs.
4. Better decision-making.
5. Increased organizational efficiency.

Features of EAI

1. Application connectivity.

ERP and E-Commerce

ERP and E-Commerce work together to automate online business operations and improve customer service.

E-Commerce systems handle online sales and customer interactions, while ERP manages inventory, finance, procurement, and order processing.

Integration of ERP and E-Commerce

1. Online orders are automatically transferred to ERP.
2. Inventory information is updated in real time.
3. Customer data is shared between systems.
4. Payment information is processed automatically.
5. Shipping and delivery status are tracked efficiently.

Benefits

1. Faster order processing.
2. Improved customer satisfaction.
3. Better inventory management.
4. Reduced manual work.
5. Increased business efficiency.

Applications

- Online retail stores.
- Digital marketplaces.
- Business-to-Business (B2B) transactions.
- Business-to-Consumer (B2C) transactions.

ERP and Internet

The Internet has transformed ERP systems by enabling global access, remote operations, and real-time communication.

Modern ERP systems use web technologies to allow users to access business information from anywhere.

Features of Internet-Based ERP

1. Web-based access.

2. Remote connectivity.
3. Online collaboration.
4. Cloud deployment.
5. Real-time data sharing.

Advantages

1. Access from any location.
2. Reduced infrastructure costs.
3. Faster communication.
4. Improved collaboration.

Future Directions in ERP

ERP systems continue to evolve with advances in technology and changing business requirements.

1) Cloud ERP: Cloud ERP systems are hosted on remote servers and accessed through the internet. **Benefits**

- Lower implementation cost.
- Easy scalability.
- Automatic updates.
- Remote accessibility.

2) Artificial Intelligence (AI): AI helps ERP systems analyze data, automate tasks, and support intelligent decision-making.

Applications

- Demand forecasting.
- Predictive maintenance.
- Fraud detection.
- Customer behavior analysis.

3) Machine Learning (ML): Machine Learning enables ERP systems to learn from historical data and improve business predictions. **Applications**

- Sales forecasting.
- Inventory optimization.
- Risk assessment.
- Process automation.

4) Internet of Things (IoT): IoT devices collect real-time data from machines, equipment, and products and integrate it with ERP systems.

Applications

- Smart manufacturing.
- Asset tracking.
- Equipment monitoring.

5. Better customer and supplier interaction.

Internet Technologies Used in ERP

1. Web browsers.
2. Cloud computing.
3. Web services.
4. APIs (Application Programming Interfaces).
5. Mobile applications.

- Supply chain visibility.

5) Mobile ERP: Mobile ERP allows employees to access ERP functions through smartphones and tablets. **Benefits**

- Anywhere access.
- Faster decision-making.
- Improved workforce productivity.
- Real-time updates.

6) Big Data Analytics: ERP systems increasingly use big data technologies to analyze large volumes of business information.

Benefits

- Better business insights.
- Improved forecasting.
- Enhanced customer understanding.
- Data-driven decision-making.

7) Social ERP: Social ERP integrates social networking and collaboration tools with ERP systems. **Benefits**

- Better communication.
- Enhanced teamwork.
- Faster information sharing.
- Improved employee engagement.

8) Blockchain Integration: Blockchain technology can improve transparency, security, and traceability within ERP systems.

Applications

- Supply chain tracking.
- Secure transactions.

- Contract management.

- Financial auditing.